

Lab 01: linear algebra + computing review

Ikea furniture

Your name + Your teammates' names

2026-08-31

Setup

Load packages and data:

```
library(tidyverse)
library(tidymodels)
ikea <- read_csv("data/ikea.csv")
```

Linear algebra review

knitr::include_graphics("exercise1.jpg")

1. a: 3×2 2. a) is ii, This is the only combination where row ^{dimensions} match with column.
 b: 2×4
 c: 2×3 b) $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \times \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 2 \cdot 2 + 3 \cdot 3 & 1 \cdot 2 + 2 \cdot 2 + 3 \cdot 2 & 1 \cdot 3 + 2 \cdot 2 + 3 \cdot 2 \\ 4 \cdot 1 + 5 \cdot 2 + 6 \cdot 3 & 4 \cdot 2 + 5 \cdot 2 + 6 \cdot 2 & 4 \cdot 3 + 5 \cdot 2 + 6 \cdot 2 \end{bmatrix} = \begin{bmatrix} 14 & 12 & 13 \\ 32 & 28 & 31 \end{bmatrix}$
 d: $m \times n \times p$
 e: $n \times n$

3. a: True, same dimensions, add all corresponding elements together.
 b: True, This is the distributive law of multiplication.
 c: False, This is a weird form of distributive law. ~~Matrix~~ Matrix multiplication is irreversible.

4. a) X is non-singular. The dimension is not $n \times n$ but $n \times p$.
 b) We don't have enough condition to determine if $X^T X$ is singular. This is because we don't know if the row or column is linearly dependent or not.

5. a) $a-b = \begin{bmatrix} a_1 - b_1 \\ \vdots \\ a_n - b_n \end{bmatrix}$ b. of $X_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $X_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, $X_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$
 We want:
 $a_1 X_1 + a_2 X_2 + a_3 X_3 = 0$ however we can't find corresponding a_1, a_2, a_3
 $\therefore$ This is a full column rank matrix.

b) $X_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $X_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, $X_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$
 We want:
 $a_1 X_1 + a_2 X_2 + a_3 X_3 = 0$. We can't find corresponding a_1, a_2, a_3 .
 $\therefore$ This is a full column rank matrix.

c) $X_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $X_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $X_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, $X_4 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$
 We want:
 $a_1 X_1 + a_2 X_2 + a_3 X_3 + a_4 X_4 = 0$.
 let $a_1 = 1, a_2 = -1, a_3 = -1, a_4 = 1$:
 $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 0$
 $\therefore$ This is not a full column rank matrix.

The answer 1e changed into P^*P .

Computing review

Exercise 7

```
ikea <- read_csv("data/ikea.csv")
```

Rows: 3694 Columns: 13

-- Column specification -----

Delimiter: ","

chr (6): name, category, link, other_colors, short_description, designer

dbl (6): X1, item_id, depth, height, width, price_usd

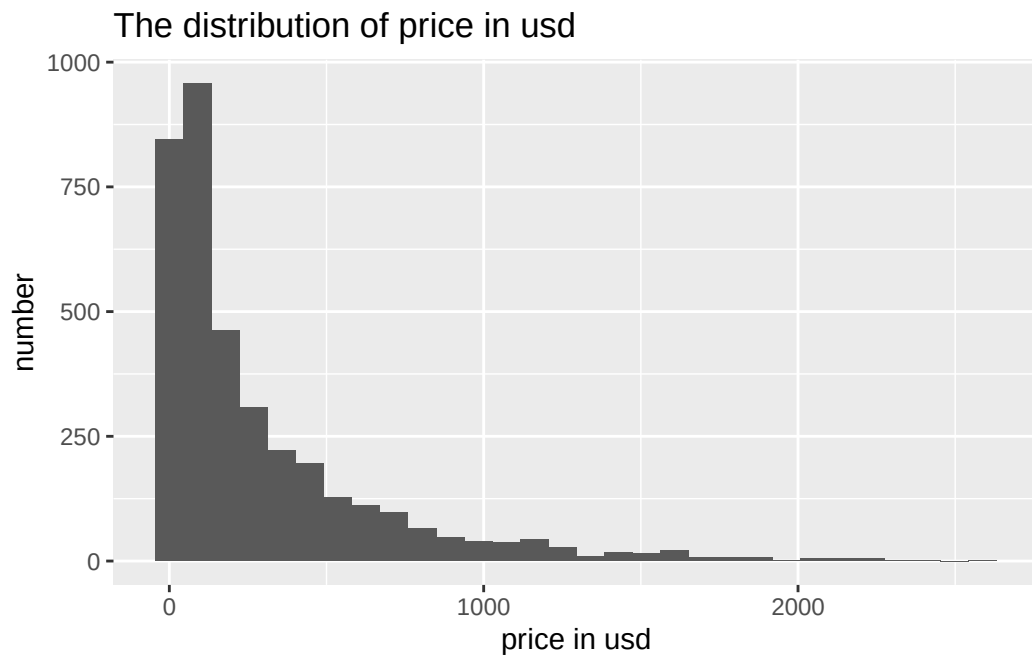
lgl (1): sellable_online

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
ggplot(data = ikea, aes(x = price_usd)) +  
  geom_histogram() +  
  labs(x = "price in usd",  
       y = "number",  
       title = "The distribution of price in usd")
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



Exercise 8

The distribution is skewed to the left, and there are some outliers that will affect the mean to be larger than the median.

Exercise 9

```
ikea_sub <- ikea |>
  filter(category %in% c( "Sofas & armchairs",
                        "Bookcases & shelving units"))

ikea_sub <- ikea_sub |>
  drop_na(width, price_usd)
```

- a. There is 765 observations in ikea_sub dataframe, and there are 13 variables.
- b. The advantage of complete case analysis is it made further operations more convenient so that there won't be any errors. A disadvantage of complete case analysis is that it might drop out too many elements so that it creates a biased result.

Exercise 10

```
ggplot(data = ikea_sub, aes(x = width, y = price_usd))+  
  geom_point() +  
  labs(x = "price in usd",  
       y = "width")
```

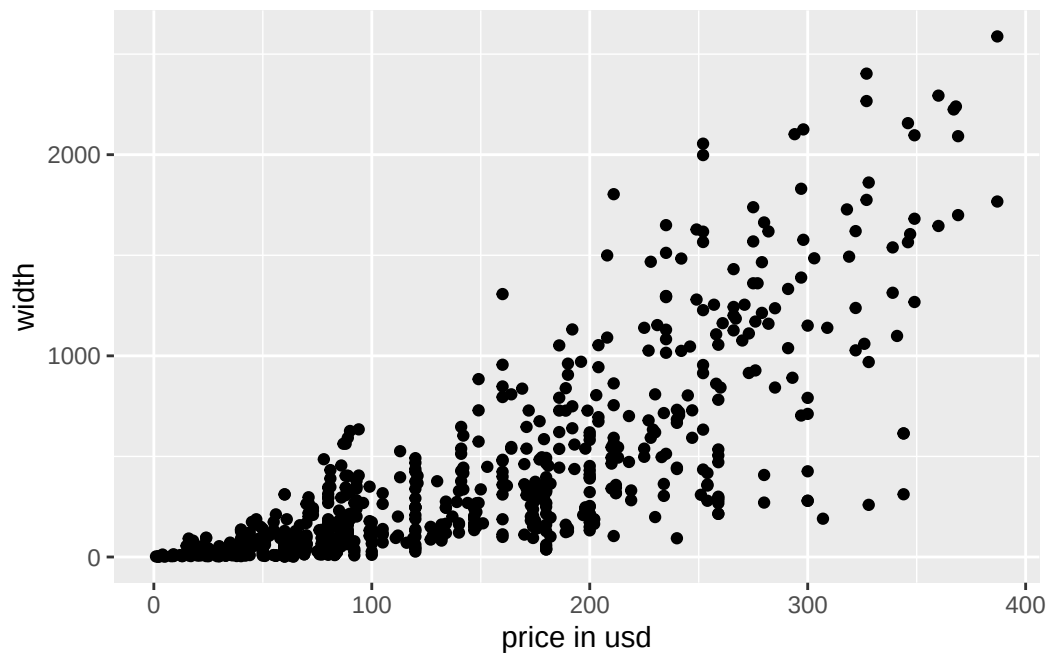


Figure 1: The dataset shows a relatively positive relationship between width and price in usd.

```
ggplot(data = ikea_sub, aes(x = width, y = price_usd, color = category, shape = category))+  
  geom_point() +  
  labs(x = "price in usd",  
       y = "width")
```

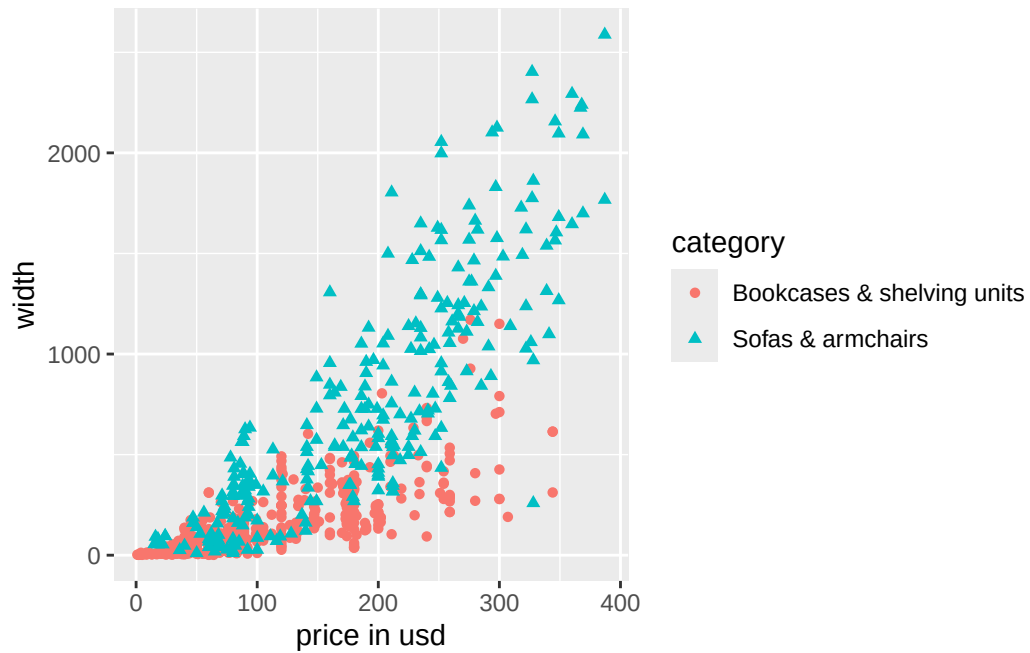


Figure 2: Both categories shows a relatively positive relationship between price in usd and width, but the sofas & armchairs category shows a steeper slope, meaning a greater change in price when the width changes.